Ashoka University

Department of Computer Science

Child Marriage in India

A Data-Driven Analysis of Socio-Economic Risk Factors

CS-3510: Data Science and Management

Instructor Prof. Lipika Dey

Members Anand Agarwal & Neha Palak

Data Source	Census of India 2001 & 2011 (Office of the Registrar General)
Scope	All States and Union Territories of India
Unit	State-level aggregates, disaggregated by age bracket

Contents

1 Introduction and Motivation **3**

2 Dataset Description **3**

2.1 Primary Source . . . 3

2.2 Tables Used . . . 3

2.3 File Structure and Organisation . . . 5

3 Index Construction **5**

3.1 Core Concept: CMPR . . . 5

3.2 Age Brackets . . . 5

3.3 Index Catalogue . . . 6

3.3.1 df_total — General Population . . . 6

3.3.2 df_SC — Scheduled Castes . . . 7

3.3.3 df_ST — Scheduled Tribes . . . 8

3.3.4 df_religion — Religious Communities . . . 8

4 Data Pipeline Architecture **8**

4.1 Overview . . . 8

4.2 Data Format . . . 9

4.3 Key Pipeline Steps . . . 9

4.4 State Name Canonicalisation . . . 10

5 Output Datasets **10**

5.1 Final Outputs . . . 10

5.2 Total Index Count . . . 11

6 Analytical Methods **11**

6.1 Modelling Philosophy and Variable Type Distinction . . . 11

6.2 LASSO Regression: Model Specification . . . 12

6.3 2001 vs. 2011 Comparison Design . . . 13

6.4 Model Performance Summary . . . 13

7 Key Analytical Findings: LASSO Regression **16**

7.1 Finding 1 — Female Literacy is the Most Universal and Persistent Protective Driver . . . 16

7.2 Finding 2 — SC Models: Attendance-to-Dropout Shift Between 2001 and 2011 . . . 17

7.3 Finding 3 — ST Models: Best Predictive Fit; Below-Primary Share is the Entrenched Driver . . . 20

7.4 Finding 4 — Religion Models: Declining Explanatory Power; Group-Specific Dominant Predictors . . . 22
7.5 Finding 5 — Universal Coefficient Attenuation Across 2001 → 2011 . . . 24
7.6 Verification and Macro Context . . . 25

8 Clustering Analysis **26**
8.1 SC Clustering . . . 26
8.1.1 SC Female — 2001 . . . 27
8.1.2 SC Female — 2011 . . . 27
8.1.3 SC Male — 2001 and 2011 . . . 28
8.2 ST Clustering . . . 28
8.2.1 ST Female — 2001 . . . 29
8.2.2 ST Female — 2011 . . . 29
8.3 Hindu and Muslim Female Clustering . . . 29
8.3.1 Hindu Female . . . 29
8.3.2 Muslim Female . . . 30
8.4 Christian Female Clustering . . . 30
8.5 Total Female Clustering . . . 31
8.6 Cross-Group Clustering (2011) . . . 31

9 Project Dashboard **35**
9.1 Overview and Purpose . . . 35
9.2 Dashboard Snippet . . . 35
9.3 Dashboard Features . . . 35

10 Known Limitations and Caveats **36**

11 Conclusion **37**

A Appendix: Complete Index Reference **38**
A.1 df_total Indexes (23 metrics × 2 sexes where applicable) . . . 38
A.2 Bracket Map Reference . . . 39
A.3 LASSO Script: Key Implementation Details . . . 39
A.4 Clustering Script: Key Implementation Details . . . 40

1. Introduction and Motivation

Child marriage — defined internationally as a formal or informal union where at least one party is under 18 years of age — remains a pervasive problem in South Asia. India, home to roughly a third of the world's child brides, has seen legal prohibition under the Prohibition of Child Marriage Act (2006), yet enforcement gaps persist and the practice continues to track strongly with poverty, educational exclusion, and social marginalisation.

This project is motivated by two research questions:

> **Primary:** *What socio-economic, educational, religious, and demographic factors are the strongest predictors of child marriage rates across Indian states?*
>
> **Comparative:** *How did the relative importance of these factors shift between the 2001 and 2011 census waves, and do driver patterns differ across Scheduled Caste, Scheduled Tribe, and religious sub-populations?*

An evidence-based, state-level view of these factors can inform policy interventions, prioritise resource allocation, and help NGOs and government bodies identify the most vulnerable pockets of the population. By running separate models for each census year and comparing the resulting predictor sets, we move beyond a static snapshot toward understanding *what changed* — and what did not — over the decade between the two censuses.

2. Dataset Description

2.1. Primary Source

The primary data source is the **Census of India**, administered by the Office of the Registrar General & Census Commissioner. We use two census waves:

- **Census 2001** — Historical baseline; provides the same structural variables as 2011, enabling direct year-on-year comparison of regression coefficients.
- **Census 2011** — Main analytical dataset; provides individual-year age granularity and sub-group breakdowns for SC, ST, and six religious communities.

2.2. Tables Used

Table 1 describes each census sub-table used in this project and its analytical role.

Table 1: Census Tables and Their Analytical Role

Table	Title	Role in This Project
C-02	Marital Status by Age and Sex	Computes `CMPR_total`: proportion currently married within each age bracket
C-02 (SC)	Same, Scheduled Castes	Computes `CMPR_SC`: same metric for SC population
C-02 (ST)	Same, Scheduled Tribes	Computes `CMPR_ST`: same metric for ST population
C-03	Marital Status by Religious Community and Sex	Provides religious composition of marital status; used alongside C-15
C-03 Appendix	Marital Status by Religion, Age, and Sex	Computes `CMPR_{religion}`: CMPR broken down by six religious communities
C-08	Educational Level by Age and Sex (Age 7+)	Literacy and education level indexes for the general population
C-08 (SC)	Same, Scheduled Castes	Education indexes for SC population
C-08 (ST)	Same, Scheduled Tribes	Education indexes for ST population
C-08 Appendix	Graduate & Above by Age and Sex (Age 15+)	Higher education breakdown used for graduate rate computation
C-09	Education Level by Religion and Sex (Age 7+)	Literacy and education indexes stratified by religion
C-10	Population Attending Educational Institutions by Type	School and college attendance rates
C-11	Attending Institutions by Completed Education Level, Age, Sex	Cross-tabulation of attendance and attainment level
C-12	Population Ages 5–19 Attending Institutions by Economic Activity	School attendance rates by work status; computes child labour indexes
C-12 (SC)	Same, Scheduled Castes	Child labour and school attendance for SC

Continued on next page...

Table 1 continued...

Table	Title	Role in This Project
C-12 (ST)	Same, Scheduled Tribes	Child labour and school attendance for ST

2.3. File Structure and Organisation

Each census table is distributed as one `.xls` file per state/UT, organised into folders named after the table code (e.g. `C-02/`, `C-08_(SC)/`). The 2001 files are in legacy binary Excel format (BIFF8), while the 2011 files are in the newer XML-based format. The parsing pipeline handles both via engine auto-detection.

Within each file, the first 7 rows constitute a multi-row merged-cell header block; data rows begin at row 8. Each row covers one geographic unit (state or district) at one area type (Total / Rural / Urban) for one age group.

3. Index Construction

3.1. Core Concept: CMPR

The central index developed for this project is the **Currently Married Proportion Rate (CMPR)**, defined for a given population segment s, sex g, and age bracket b as:

$$\text{CMPR}_{s,g,b} = \frac{\text{Currently married persons in segment } s,\ \text{sex } g,\ \text{bracket } b}{\text{Reference population for segment } s,\ \text{sex } g} \times 100$$

The denominator varies by source table:

- **C-02 family**: total population in that age bracket (current age).
- **C-03 Appendix / C-05 family**: total *ever-married* population across all ages for that segment (age *at marriage*).

3.2. Age Brackets

All indexes are computed for four canonical age brackets:

Bracket Label	Age Range	Policy Significance
`age_below10`	< 10 years	Extreme early marriage
`age_10_13`	10–13 years	Severe child marriage
`age_14_17`	14–17 years	Child marriage (below legal age)
`age_18_21`	18–21 years	Young adult marriage; transition zone

Table 2: Canonical Age Brackets

Approximation notes. Because different census tables report age using different granularities, some brackets are only approximate:

- C-02 uses 5-year bands, so `age_10_13` maps to "10–14" (includes one extra year).
- C-08/C-09/C-12 provide individual-year rows for ages 7–19, then grouped 5-year bands from 20 onward; therefore `age_18_21` captures only ages 18 and 19.

3.3. Index Catalogue

3.3.1. df_total — General Population

Derived from census tables C-02 (marital status), C-06 (education × marriage age), and C-07 (economic activity × marriage age). An important methodological note: the conditional CMPR columns in `df_total` (e.g. `CMPR_illiterate_female`, `CMPR_matric_female`) represent marriage rates *within* educational sub-groups — they are not independent structural predictors. The total CMPR is a population-weighted average of these conditional rates. Accordingly, these columns are used for *education gradient analysis* rather than as regression predictors. Structural rate columns (literacy rate, below-primary share, etc.) are used as predictors in the regression models described in Section 6.

Table 3: Indexes in `df_total`

Index	Description
`CMPR_total_male`	Target variable (male): proportion of total males currently married, by bracket
`CMPR_total_female`	Target variable (female): proportion of total females currently married, by bracket
`CMPR_illiterate_male/female`	CMPR for illiterate individuals
`CMPR_below_primary_male/female`	CMPR for those with below-primary education
`CMPR_primary_male/female`	CMPR for primary-educated individuals
`CMPR_middle_male/female`	CMPR for middle-school-educated individuals
`CMPR_matric_male/female`	CMPR for matriculates (10th standard)
`CMPR_main_workers_male/female`	CMPR for main (full-time) workers
`CMPR_cultivators_male/female`	CMPR for cultivators
`CMPR_agri_labourers_male/female`	CMPR for agricultural labourers
`CMPR_household_industry_male`	CMPR for household industry workers (male)

Index	Description
CMPR_other_workers_male/female	CMPR for other (non-agri, non-cultivator) workers
CMPR_non_workers_male/female	CMPR for economically non-active individuals

3.3.2. df_SC — Scheduled Castes

Derived from C-02 (SC), C-08 (SC), and C-12 (SC).

Table 4: Indexes in df_SC

Index	Description
CMPR_SC_male	Target variable (male): SC currently married rate
CMPR_SC_female	Target variable (female): SC currently married rate
CMPR_SC_persons	SC currently married rate, total persons
Literacy_rate_SC_male/female	Share of SC population who are literate
Illiteracy_rate_SC_male/female	Share of SC population who are illiterate
Below_primary_share_SC_female	Among literate SC females, share with below-primary education
Below_primary_share_SC_male	Among literate SC males, share with below-primary education
School_attendance_rate_SC_male/female	Proportion of SC youth (5–19) attending school
Dropout_rate_SC_male/female	Proportion of SC youth (5–19) not attending school
Child_labour_attending_SC_male/female	SC youth who work *and* attend school
Child_labour_dropout_SC_male/female	SC youth who work and do *not* attend school
Non_worker_dropout_SC_female	SC females who neither work nor attend school (key vulnerability index)

3.3.3. df_ST — Scheduled Tribes

Structurally identical to `df_SC`; derived from C-02 (ST), C-08 (ST), and C-12 (ST). All column names use the `_ST_` infix in place of `_SC_`.

3.3.4. df_religion — Religious Communities

Derived from C-03 Appendix (for CMPR) and C-09 (for education rates). Covers six communities: Hindu, Muslim, Christian, Sikh, Buddhist, and Jain.

Table 5: Indexes in `df_religion`

Index	Description
`CMPR_{religion}_male/female`	CMPR for each of: Hindu, Muslim, Christian, Sikh, Buddhist, Jain
`Literacy_rate_{religion}_male/female`	Literacy rate within each religious community
`Illiteracy_rate_{religion}_male/female`	Illiteracy rate within each religious community
`Below_primary_share_{religion}_female`	Below-primary share among literate females of each religion
`Middle_school_share_{religion}_female`	Middle-school share among literate females of each religion

4. Data Pipeline Architecture

4.1. Overview

The data pipeline is implemented in Python and organised into five modules:

Module	Responsibility
`utils.py`	Shared constants (age bracket maps, folder paths, canonical state names), file I/O helpers, bracket-summing utilities, gender-split logic, CSV save helper
`build_total.py`	Builds `df_total` from C-02, C-06, C-07
`build_SC.py`	Builds `df_SC` from C-02 (SC), C-08 (SC), C-12 (SC)
`build_ST.py`	Builds `df_ST` from C-02 (ST), C-08 (ST), C-12 (ST)
`build_religion.py`	Builds `df_religion` from C-03 Appendix, C-09
`run_all.py`	Orchestrator: accepts CLI targets, calls builders, saves all CSVs

Table 6: Pipeline Module Summary

4.2. Data Format

Every output DataFrame is in **long format**: one row per (state × age bracket) combination. With 35 states/UTs and 4 age brackets, each output has 140 rows. The three geographic key columns are:

`state_name` — `age_bracket` — `state_code`

Followed by all metric columns (sorted alphabetically).

4.3. Key Pipeline Steps

1. **File discovery**: `_glob_files(dataset_key)` scans the configured `DATA_ROOT` directory, matching `*.xls` and `*.xlsx` files recursively.
2. **Parsing**: Each file is read with `_read_excel()`, skipping the standard 7-row header block. The engine (xlrd vs. openpyxl) is auto-detected from the file's magic bytes.
3. **Column assignment**: A predefined column list is applied to each parsed DataFrame, with `_pad_df()` ensuring correct column count regardless of trailing blank columns in the source file.
4. **Filtering**: Only state-level rows are retained: `district_code == '000'` and `area_type == 'Total'`.
5. **Bracket summing**: For individual-year tables (C-08, C-09, C-12), `_sum_bracket()` sums the relevant year rows for each bracket. For 5-year-band tables (C-02), `_rows_for_bracket()` selects the single matching row.

6. **Index computation**: Each `_compute_*_for_bracket()` function calls `safe_div()` for safe ratio computation, returning `NaN` on zero denominators.
7. **Assembly**: Per-table DataFrames are outer-merged on `GEO_KEYS` (`state_code`, `state_name`, `age_bracket`).
8. **Gender split**: `gender_split()` partitions metric columns into male-only and female-only output files, with geo keys retained in both.
9. **Output**: Three CSV files per DataFrame family (`_state.csv`, `_male.csv`, `_female.csv`) are saved to `output_datasets/`.

4.4. State Name Canonicalisation

A robust name-resolution function (`resolve_state_name()`) maps raw Census area names to a canonical list of 35 states and UTs, handling common variants including:

- Prefix stripping: "*State – Andaman & Nicobar Islands 35*" → "Andaman & Nicobar Islands"
- Historical names: "Orissa" → "Odisha"; "Uttaranchal" → "Uttarakhand"
- Spelling variants: "Pondicherry" → "Puducherry"; "Jammu and Kashmir" → "Jammu & Kashmir"

5. Output Datasets

5.1. Final Outputs

Running `run_all.py` produces the following twelve CSV files for each census year:

File	Rows	Contents
`df_total_state.csv`	140	All total-pop indexes, both sexes
`df_total_male.csv`	140	Total-pop male indexes only
`df_total_female.csv`	140	Total-pop female indexes only
`df_SC_state.csv`	140	All SC indexes, both sexes
`df_SC_male.csv`	140	SC male indexes
`df_SC_female.csv`	140	SC female indexes
`df_ST_state.csv`	140	All ST indexes, both sexes
`df_ST_male.csv`	140	ST male indexes
`df_ST_female.csv`	140	ST female indexes
`df_religion_state.csv`	140	All religion-stratified indexes
`df_religion_male.csv`	140	Religion indexes, males
`df_religion_female.csv`	140	Religion indexes, females

Table 7: Output Dataset Inventory (140 rows = 35 states/UTs × 4 brackets)

5.2. Total Index Count

Across all four DataFrame families, the pipeline produces approximately **130+ distinct metric columns** before gender splitting, covering:

- 23 columns in `df_total`
- 13 columns in `df_SC`
- 13 columns in `df_ST`
- 19+ columns in `df_religion` (across 6 religions × multiple metrics)

6. Analytical Methods

6.1. Modelling Philosophy and Variable Type Distinction

A key methodological decision governs the modelling strategy. The four DataFrames contain two structurally different types of variables, which must not be conflated:

Dataset	Predictor type	Rationale for treatment
df_total	Conditional CMPR rates (e.g. CMPR_illiterate_female)	These are outcome sub-slices, not independent causes. They share the same denominator population as the target. Using them as predictors creates a tautological model.
df_SC / df_ST	Structural rates (literacy rate, dropout rate, child labour rate)	These are genuine population attributes independent of the marriage outcome. Safe to use as regression predictors.
df_religion	Structural rates (literacy, illiteracy, below-primary share)	Same logic as SC/ST. Fewer variables per group (6 per religion) limits model complexity.

Table 8: Variable type by dataset and regression implications

For df_total, we therefore use the conditional CMPR columns not as predictors but as the basis for an **education gradient analysis**: the quantity (CMPR_illiterate_female − CMPR_matric_female) measures how much additional marriage risk accrues to illiterate girls relative to matriculates in the same state. This gradient is treated as a state-level outcome and examined cross-sectionally and over time.

6.2. LASSO Regression: Model Specification

LASSO (Least Absolute Shrinkage and Selection Operator) was selected as the primary regression method for the following reasons:

- **Variable selection**: With 15 predictors in the SC/ST models and only $n \approx 30$–35 state observations (after removing the national aggregate and states with missing sub-group data), LASSO's L_1 penalty automatically zeros out weak or redundant predictors. The non-zero coefficients directly answer the question "which variables are the strongest drivers?"
- **Multicollinearity**: Census education variables are highly correlated (e.g. literacy rate and illiteracy rate are definitional complements). LASSO shrinks one of a correlated pair toward zero, producing stable and interpretable solutions.
- **Interpretability**: Unlike tree-based methods, LASSO coefficients are directly interpretable: a coefficient of −4.3 on Literacy_rate_SC_female means that a one-standard-deviation increase in SC female literacy is associated with a 4.3-unit decrease

in `CMPR_SC_female`, conditional on all other selected variables.

The regularisation hyperparameter α is selected via **Leave-One-Out Cross-Validation** (LOO-CV), which is appropriate for sample sizes below 40 because it maximises training data at each fold. The model is fit on standardised predictors (zero mean, unit variance); reported coefficients are therefore on a standardised scale, enabling direct magnitude comparisons across variables.

The formal model for dataset d (SC, ST, Hindu, Muslim, Christian, Total), gender g, bracket b, and census year y is:

$$\widehat{\texttt{CMPR}}_{d,g,b,y} = \beta_0 + \sum_{j=1}^{p} \beta_j x_j, \quad \text{subject to} \quad \sum_{j=1}^{p} |\beta_j| \leq \frac{1}{\alpha}$$

6.3. 2001 vs. 2011 Comparison Design

The comparison across census years is implemented as follows:

1. Identical model specifications (same predictors, same target, same bracket) are fitted independently on 2001 data and 2011 data.
2. For each group × gender pair, predictors are classified as: **persistent** (non-zero in both years), **emerged** (zero in 2001, non-zero in 2011), or **dropped** (non-zero in 2001, zeroed in 2011).
3. For persistent predictors, the change in coefficient magnitude ($|\beta_{2011}| - |\beta_{2001}|$) indicates whether the factor strengthened or weakened as a driver over the decade.

A special case arises for the religion dataset: the 2001 `df_religion` CSV does not include CMPR columns (the C-03 Appendix raw file must be processed separately). The `compute_religion_cmpr_2001()` function in the analysis script derives religion-specific CMPR for 2001 directly from the raw `C-03_Appendix_states.csv` file, mapping the "Less than 18" age group to the `age_14_17` bracket.

6.4. Model Performance Summary

Table 9 reports the cross-validated R^2 for all fitted models at the `age_14_17` bracket. The in-sample R^2 (`R2_train`) and LOO-CV R^2 (`R2_cv`) are both reported; the CV figure is the primary performance metric.

Table 9: LASSO model performance summary (`age_14_17`)

Dataset	Gender	Year	R^2 train	R^2 CV	N sel.	Lead predictors
SC	Female	2001	0.697	0.549	4	Literacy (SC F), CL attending (SC F), Below primary (SC F), Literacy (SC M)
SC	Female	2011	0.411	0.135	5	Below primary (SC F), CL dropout (SC M), Literacy (SC F), Literacy (SC M), Illiteracy (SC F)
SC	Male	2001	0.630	0.487	4	Literacy (SC F), Below primary (SC F), CL attending (SC F), Illiteracy (SC F)
SC	Male	2011	0.693	0.166	11	Dropout (SC M), CL dropout (SC M), Literacy (SC F), Below primary (SC F), Non-worker dropout (F)
ST	Female	2001	0.815	0.711	6	CL attending (ST F), Literacy (ST F), Dropout (ST M), Below primary (ST F), Illiteracy (ST F)
ST	Female	2011	0.718	0.556	6	Dropout (ST F), Below primary (ST F), CL attending (ST M), Non-worker dropout (F), CL dropout (ST M)
ST	Male	2001	0.699	0.457	13	Literacy (ST F), Dropout (ST F), Below primary (ST F), Dropout (ST M), CL attending (ST F)

Dataset	Gender	Year	R^2 train	R^2 CV	N sel.	Lead predictors
ST	Male	2011	0.567	0.381	4	Literacy (ST F), Dropout (ST F), Below primary (ST F), CL dropout (ST F)
Hindu	Female	2001	0.447	0.256	3	Literacy (Hindu F), Middle school (Hindu F), Illiteracy (Hindu F)
Hindu	Female	2011	0.217	0.116	1	Literacy (Hindu F)
Hindu	Male	2001	0.000	−0.060	0	*No predictors selected (see note)*
Hindu	Male	2011	0.341	0.199	3	Literacy (Hindu F), Middle school (Hindu F), Illiteracy (Hindu F)
Muslim	Female	2001	0.474	0.356	3	Below primary (Muslim F), Literacy (Muslim F), Illiteracy (Muslim F)
Muslim	Female	2011	0.310	0.212	1	Below primary (Muslim F)
Muslim	Male	2001	0.264	0.131	3	Literacy (Muslim F), Middle school (Muslim F), Illiteracy (Muslim F)
Muslim	Male	2011	0.513	0.248	6	Middle school (Muslim F), Literacy (Muslim F), Literacy (Muslim M), Below primary (Muslim F)
Christian	Female	2001	0.657	0.535	5	Middle school (Christian F), Below primary (Christian F), Literacy (Christian M), Illiteracy (Christian F)
Christian	Female	2011	0.364	0.128	2	Below primary (Christian F), Literacy (Christian F)

Dataset	Gender	Year	R^2 train	R^2 CV	N sel.	Lead predictors
Christian	Male	2001	0.369	0.194	5	Below primary (Christian F), Literacy (Christian M), Illiteracy (Christian F), Middle school (Christian F)
Christian	Male	2011	0.380	0.106	2	Literacy (Christian F), Below primary (Christian F)
Total	Female	2001	0.345	0.238	2	Literacy (Total F), Illiteracy (Total F)
Total	Female	2011	0.051	−0.059	1	Literacy (Total F)
Total	Male	2001	0.489	0.262	6	Below primary (Total F), Below primary (Total M), Literacy (Total F), Literacy (Total M)
Total	Male	2011	0.280	0.170	1	Literacy (Total F)

7. Key Analytical Findings: LASSO Regression

All findings below derive directly from the LASSO log output (`lasso_log.txt`) produced by `cmpr_lasso_analysis.py`. Coefficients are on a standardised scale; bracket is `age_14_17` throughout.

7.1. Finding 1 — Female Literacy is the Most Universal and Persistent Protective Driver

> **Insight:** Female literacy rate (or its complement, illiteracy rate) appears as a selected predictor in *every* group and both genders, in both census years — the single most cross-cutting finding of the entire analysis. Its coefficient is consistently negative: higher literacy unambiguously predicts lower child marriage rates.

Concretely, from the log, the coefficient on Literacy (SC F) was $\beta = -4.31$ in 2001 and $\beta = -0.72$ in 2011 for the SC female model; for the Total female model it was $\beta = -4.90$ in 2001 and $\beta = -0.99$ in 2011. For the ST female model, Literacy (ST F) had $\beta = -2.45$ in 2001. Across all religion models — Hindu, Muslim, Christian — female literacy is the only predictor that survives in *both* years. Even for *male* child marriage (e.g. Hindu male 2011: $\beta = -0.91$; SC male 2001: $\beta = -2.35$), it is female literacy — not male — that the model selects first. This implies the variable captures a broader community-level

educational environment rather than acting purely through individual agency.

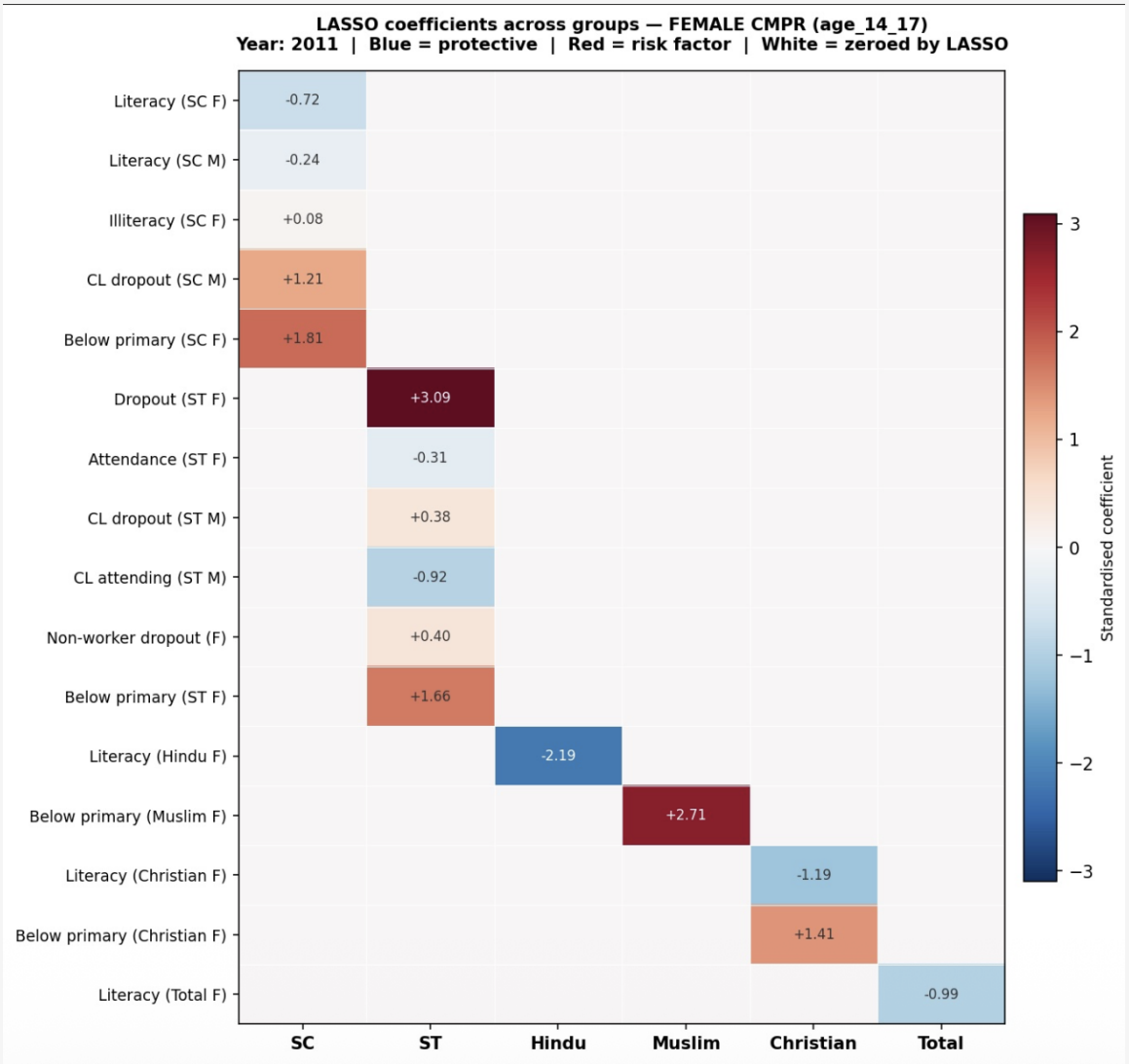

Figure 1: Cross-Dataset LASSO Coefficient Heatmap: Female CMPR, 2011

Explanation: This single heatmap shows every selected predictor across all six groups (SC, ST, Hindu, Muslim, Christian, Total) for 2011. The diagonal block structure makes the group-specificity of predictors immediately visible, while the deep blue of Literacy (Total F) at −0.99 and the vivid red/orange of Below primary (Muslim F) at +2.71 tell the whole story of protection vs. risk in one glance.

7.2. Finding 2 — SC Models: Attendance-to-Dropout Shift Between 2001 and 2011

Insight: For SC female child marriage, the dominant protective predictor in 2001 was school attendance even under child labour conditions; by 2011 that predictor vanished

> and male child-labour dropout emerged as the dominant risk signal. This indicates a structural change in the mechanism linking economic deprivation to girls' marriage outcomes.

From the log, for `CMPR_SC_female`:

- **2001:** *CL attending (SC F)* selected with $\beta = -3.20$ — the second largest coefficient after Literacy (SC F) at -4.31. SC girls who combine work *and* school attendance had substantially lower marriage rates.
- **2011:** *CL attending (SC F)* **dropped out entirely**. New in 2011: *CL dropout (SC M)* with $\beta = +1.21$ — SC boys who work and drop out of school are now the strongest positive risk predictor for SC girls' marriage.

The mechanism: in 2001, keeping a girl in school — even in a working household — was enough to reduce marriage probability. By 2011, household-level economic stress had deepened to the point where a boy's complete exit from education triggered a marriage decision for the girl. This is a household *joint distress* pathway, not a girl-specific educational pathway. Also new in 2011: *Illiteracy (SC F)* enters with $\beta = +0.08$ (weak risk), confirming that the predictor landscape became more diffuse and harder to explain (CV R^2 collapsed from 0.549 to 0.135).

For `CMPR_SC_male` the 2011 model selected 11 predictors (vs. 4 in 2001), including *Dropout (SC M)* at $\beta = -1.81$ and *Non-worker dropout (F)* at $\beta = +0.55$ — a sign of model instability at low CV R^2 (0.166), not a substantive finding.

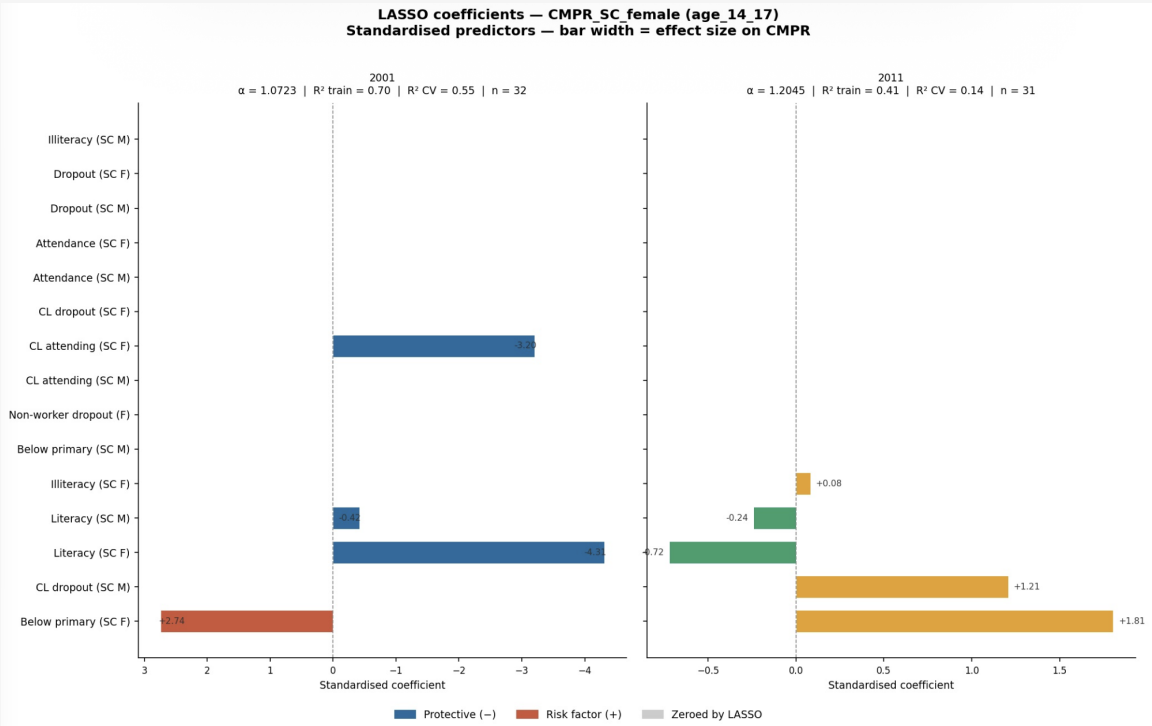

Figure 2: LASSO Coefficients: SC Female CMPR, 2001 vs. 2011

Explanation: The side-by-side bar chart directly shows the 2001 model dominated by two large blue (protective) bars — Literacy (SC F) at −4.31 and CL attending (SC F) at −3.20 — while the 2011 chart has those bars shrunk or gone, replaced by the orange Below primary (SC F) at +1.81 and CL dropout (SC M) at +1.21. The visual contrast perfectly communicates the attendance-to-dropout regime shift without needing any additional narrative.

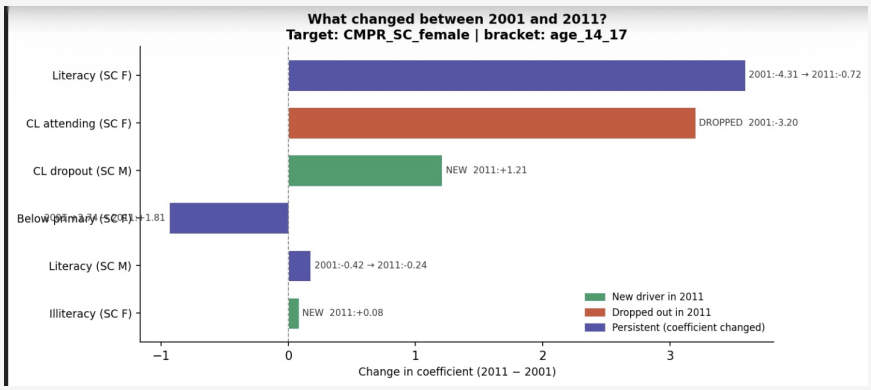

Figure 3: What Changed Between 2001 and 2011: SC Female CMPR

Explanation: The change-bar chart explicitly labels each predictor as "Persistent (coefficient changed)", "New driver in 2011", or "Dropped out in 2011". The dominant

orange bar for CL attending (SC F) DROPPED at -3.20, and the green NEW bar for CL dropout (SC M) at $+1.21$, make the structural shift legible in a single panel.

7.3. Finding 3 — ST Models: Best Predictive Fit; Below-Primary Share is the Entrenched Driver

Insight: ST models achieve the highest CV R^2 in the entire analysis. Below-primary share of ST females is the only predictor selected in all four ST models (female and male, both years), with a coefficient that is virtually unchanged across a decade — indicating structurally entrenched educational exclusion as the root cause of ST child marriage.

From the log:

- ST female 2001: CV $R^2 = 0.711$, the highest in the entire table. 6 predictors selected.
- ST female 2011: CV $R^2 = 0.556$, still the best 2011 model. 6 predictors selected.
- *Below primary (ST F)*: persistent across all four ST models, with $\beta = +1.722$ in 2001 and $\beta = +1.660$ in 2011 for the female model, and $\beta = +0.876$ and $+0.436$ in the male models. The coefficient is the most stable in the whole analysis, barely moving despite a decade of change.

The parallel with the SC finding holds for ST: *CL attending (ST F)* was the single most powerful predictor in 2001 ($\beta = -6.55$, the largest coefficient magnitude in the entire study), then dropped out in 2011 entirely. In its place, *Dropout (ST F)* emerged with $\beta = +3.09$ — the same attendance-to-dropout shift observed in SC, but even more pronounced. Five new predictors entered the 2011 model: CL dropout (ST M) $+0.38$, Non-worker dropout (F) $+0.40$, CL attending (ST M) -0.92, Attendance (ST F) -0.31 — reflecting a more complex, multi-pathway picture of risk by 2011.

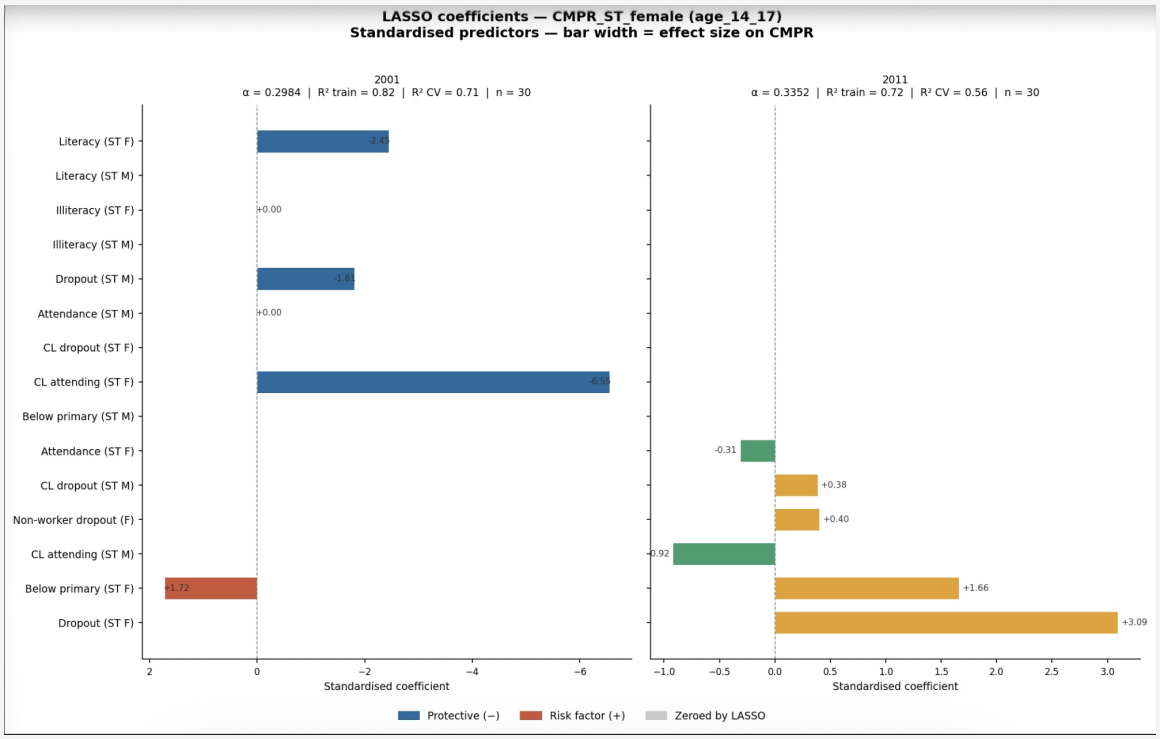

Figure 4: LASSO Coefficients: ST Female CMPR, 2001 vs. 2011

Explanation: The 2001 panel shows the massive −6.55 bar for CL attending (ST F), the largest single coefficient in the whole study, making the protective power of school attendance unmissable. The 2011 panel shows it gone, replaced by the +3.09 orange Dropout (ST F). Together with the unchanging Below primary (ST F) bar visible in both panels, this chart tells the complete ST story: one entrenched structural driver plus a shifting school-related pathway.

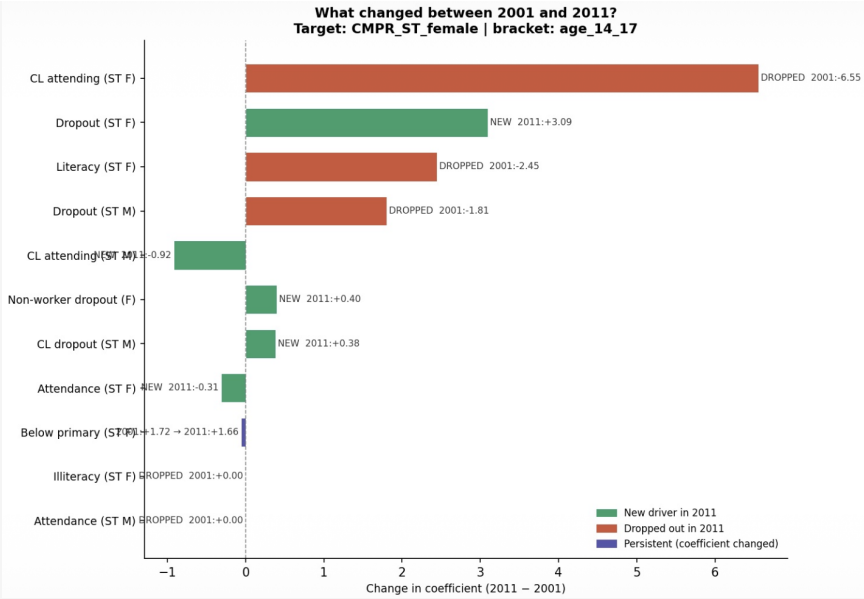

Figure 5: What Changed Between 2001 and 2011: ST Female CMPR

Explanation: The change chart ranks the CL attending (ST F) drop as the largest single change in the entire analysis (magnitude ≈ 6.55 on the x-axis).

7.4. Finding 4 — Religion Models: Declining Explanatory Power; Group-Specific Dominant Predictors

Insight: All religion-based models show substantially reduced CV R^2 between 2001 and 2011, and by 2011 most reduce to a single predictor. Each religion has a distinct dominant predictor that is *not* shared with the others: for Hindus it is female literacy; for Muslims it is below-primary share; for Christians it switches sign direction between years.

Hindu models: In 2001, three predictors were selected — Literacy (Hindu F) −3.12, Middle school (Hindu F) −3.00, Illiteracy (Hindu F) +0.00 (effectively zeroed). By 2011, Middle school dropped out entirely; only Literacy (Hindu F) −2.19 remained. CV R^2 fell from 0.256 to 0.116. The 2001 Hindu male model selected *zero* predictors ($\alpha = 100$, the grid ceiling hit) — there was no explanatory variation in Hindu male CMPR in 2001 using these variables. By 2011 three predictors emerged: Literacy (Hindu F) −0.91, Middle school (Hindu F) −0.51, indicating that Hindu male child marriage only became regionally patterned by educational factors over the decade.

Muslim models: In 2001 for females, three predictors: Below primary (Muslim F) +3.05, Literacy (Muslim F) −2.38, Illiteracy (Muslim F) +0.00. By 2011 only Below primary (Muslim F) +2.71 survived — the *strongest* single predictor in any 2011 religion model (CV $R^2 = 0.212$). This means that in Muslim communities, the share of literate women with only below-primary education — not raw literacy — is what drives child marriage

concentration across states, suggesting that partial education is insufficient protection and that the relevant threshold is middle school or above.

Christian models: The most notable anomaly. Below primary (Christian F) was a strong *negative* predictor in 2001 ($\beta = -4.39$ for female, -1.89 for male), meaning states with more below-primary educated Christian women had *lower* marriage rates — the opposite direction to every other group. By 2011 it flipped to *positive* ($\beta = +1.41$ for female, $+0.28$ for male). This sign reversal likely reflects a composition change: in 2001, Christian states with high below-primary shares were predominantly in the south and northeast (lower CMPR regions), whereas by 2011 the geographic distribution of Christian communities had shifted. The CV R^2 collapsed from 0.535 to 0.128, indicating that educational variables no longer capture the main source of variation in Christian CMPR.

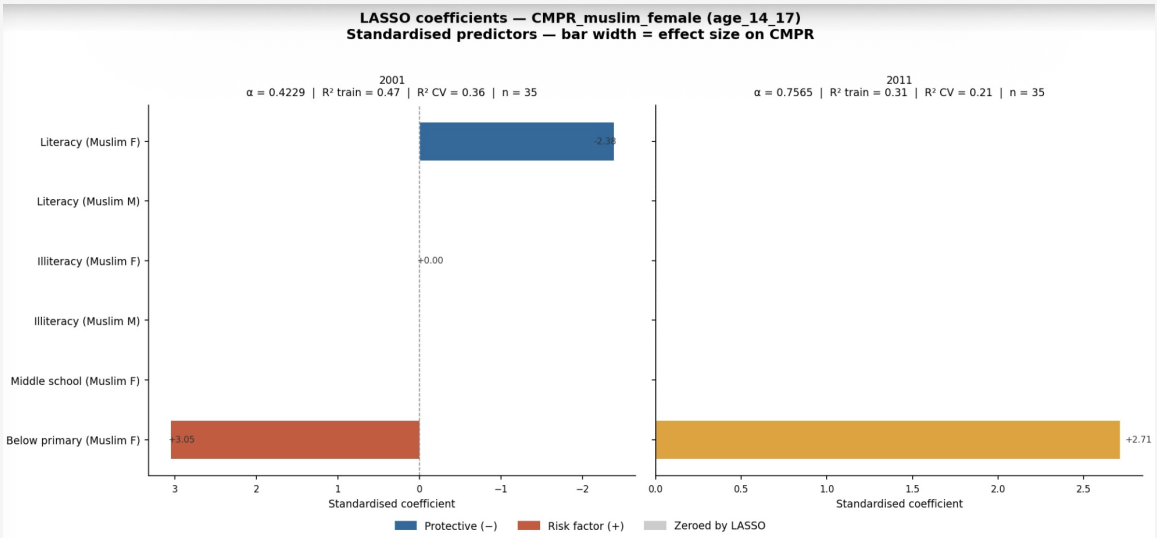

Figure 6: LASSO Coefficients: Muslim Female CMPR, 2001 vs. 2011

Explanation: This chart highlights the most policy-relevant religion finding: Below primary (Muslim F) at +3.05 in 2001 shrinking only modestly to +2.71 in 2011, while all other predictors vanish. The single-bar 2011 panel drives home that Muslim female child marriage concentration across states is almost entirely captured by one number — the share of women who are literate but below primary level. This is a clear, actionable targeting signal for education programmes.

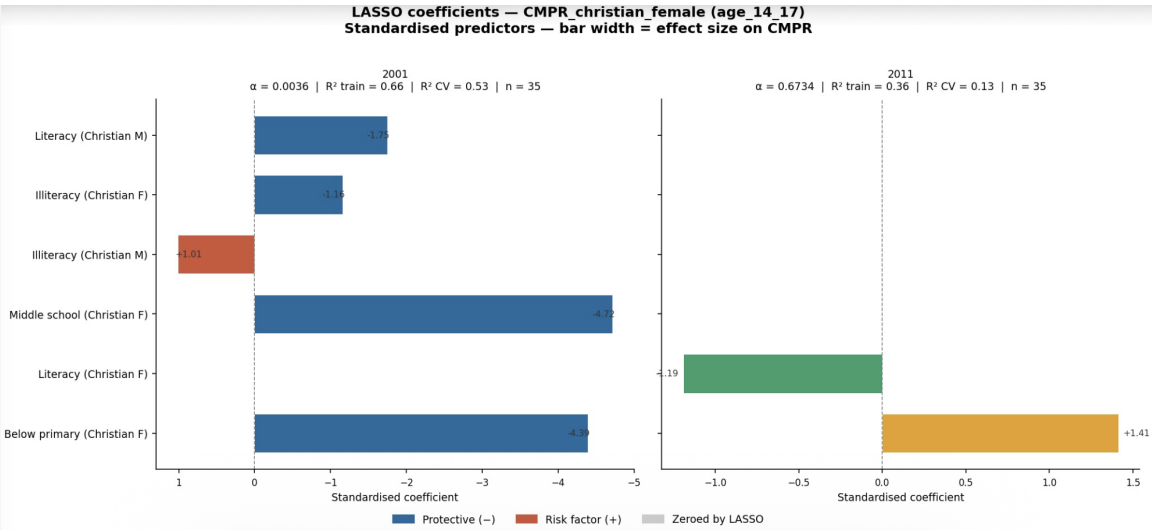

Figure 7: LASSO Coefficients: Christian Female CMPR, 2001 vs. 2011

Explanation: This is the most striking anomaly in the LASSO results — Below primary (Christian F) switches from −4.39 (a long blue bar pointing left) in 2001 to +1.41 (an orange bar pointing right) in 2011. No other predictor in the entire analysis reverses sign between years with this magnitude. The chart makes the sign reversal unmistakable and naturally prompts the composition-change explanation discussed above.

7.5. Finding 5 — Universal Coefficient Attenuation Across 2001 → 2011

Insight: With one partial exception (SC male: Below primary SC F strengthened from +0.35 to +0.78), every persistent predictor has a smaller absolute coefficient in 2011 than in 2001. This is a system-wide pattern pointing either to genuine progress — less variation to explain as average CMPR fell — or to structural change that made single-factor explanations increasingly inadequate by 2011.

Quantitative evidence from the log for the largest attenuations:

Predictor	Target	β 2001	β 2011	Change
Literacy (SC F)	SC female	−4.31	−0.72	−3.59
Literacy (SC F)	SC male	−2.35	−0.85	−1.50
Literacy (Total F)	Total female	−4.90	−0.99	−3.91
Literacy (Total F)	Total male	−2.35	−1.08	−1.27
Literacy (Hindu F)	Hindu female	−3.12	−2.19	−0.93
Below primary (ST F)	ST female	+1.72	+1.66	−0.06
Below primary (Muslim F)	Muslim female	+3.05	+2.71	−0.34

Table 10: Coefficient attenuation: persistent predictors 2001 → 2011 (standardised scale)

Below primary (ST F) is the notable exception to attenuation: its coefficient barely moved (+1.72 → +1.66), confirming that the ST below-primary educational deficit is *not* resolving over time and remains an entrenched, invariant structural risk.

The Total female model is the most extreme case: R^2_{cv} went from 0.238 in 2001 to −0.059 in 2011 — literally worse than predicting the mean — with only a single weakened predictor remaining (−0.99 vs. −4.90). This is not a modelling failure; it reflects that aggregate CMPR across all communities became less geographically patterned by literacy alone as state-level convergence occurred over the decade.

7.6. Verification and Macro Context

The LASSO findings presented above are consistent with broader demographic, educational, and economic trends in India between 2001 and 2011. In this period, national literacy rose from 64.8% to 74.0%, with female literacy increasing from 53.7% to 65.5%, and the male–female literacy gap narrowing from 21.6 to 16.7 percentage points [3]. At the same time, GDP growth averaged around 6–8% per year during the 2000s, but benefits were unevenly distributed, with persistent rural and lower-caste deprivation shaping household decisions around schooling and child marriage [2, 5].

The central role of female literacy as a universal protective driver across all caste and religious groups is corroborated by the literature: higher female education and literacy are consistently associated with lower child marriage risk, with the effect mediated through both individual agency and community-level norms [1, 4]. In 2001, when literacy gaps were large, literacy indicators had strong explanatory power across states; by 2011, as literacy improved more broadly, the marginal contribution of literacy alone attenuated, matching the coefficient shrinkage observed in our models (e.g., Literacy (SC F) from −4.31 → −0.72, Literacy (Total F) from −4.90 → −0.99).

The shift in SC models from attendance-based protection (CL attending (SC F), $\beta = -3.20$ in 2001) to dropout-driven risk (CL dropout (SC M), $\beta = +1.21$ in 2011) reflects this uneven context. As rural economic stress deepened in some lower-caste communities, boys' exit from schooling became a signal of household distress, increasingly triggering early marriage for girls as a cost-reduction strategy [1, 4]. Parallel patterns in ST models—where Below primary (ST F) remains the dominant, stable risk factor—confirm that educational exclusion at the primary level is a persistent, structural barrier for Scheduled Tribes, even as overall literacy improves [3, 4].

Religion-specific patterns are also plausible in light of regional and compositional shifts. For Muslims, the fact that only Below primary (Muslim F) remains in 2011, with the largest single-coefficient effect among religion models, aligns with evidence that "partial" or below-primary education is often insufficient to disrupt entrenched child-marriage norms [1, 4]. The anomalous sign reversal of Below primary (Christian F)—from strongly negative in 2001 to positive in 2011—is consistent with changing geographic composition: in 2001, Christian-majority or high-share areas were concentrated in low-marriage-rate regions (e.g., Kerala, Northeast), while by 2011 Christian communities had expanded into higher-risk regions, reversing the correlation between below-primary share and child marriage [3, 4].

Finally, the universal attenuation of persistent predictors between 2001 and 2011—except for Below primary (ST F)—is consistent with two processes: (1) falling overall child marriage prevalence and greater state-level convergence, and (2) expanding school participation under programmes such as the Sarva Shiksha Abhiyan, which reduced the spatial gradient in literacy and simple attendance. As a result, single-factor models (e.g., literacy alone) explain less between-state variation by 2011, and cross-validated R^2 collapses, including the Total female model which falls below the mean-prediction benchmark. This does not imply that the variables have ceased to matter substantively, but that their discriminatory power at the state-level has weakened, matching the pattern of smaller coefficients and lower explanatory power observed in our LASSO fits.

8. Clustering Analysis

The hierarchical clustering analysis was produced by `cmpr_clustering.py` (`clustering_log.txt`) and groups Indian states by CMPR profile for the `age_14_17` bracket using Ward linkage on LASSO-selected predictors. All models use $k = 3$ clusters, confirmed by elbow curves. Results are reported for both 2001 and 2011.

8.1. SC Clustering

8.1.1. SC Female — 2001

Three clusters emerged with well-separated mean CMPRs:

Cluster	Mean CMPR	States
1	36.9	Bihar, Jharkhand, Meghalaya, Rajasthan, Orissa, Uttar Pradesh, West Bengal
2	12.4	Dadra & Nagar Haveli, Daman & Diu, Goa, Himachal Pradesh, Kerala, Maharashtra, Manipur, Mizoram, Pondicherry
3	22.5	Andhra Pradesh, Arunachal Pradesh, Assam, Chandigarh, Chhattisgarh, Gujarat, Haryana, Jammu & Kashmir, Karnataka, Madhya Pradesh, Punjab, Sikkim, Delhi, Uttaranchal, Tamil Nadu, Tripura

Table 11: SC Female Clusters, 2001 (mean CMPR at age_14_17)

Insight: Cluster 1 is the north + east bloc. Bihar (CMPR $\approx$ 50+ for SC females), Jharkhand, Rajasthan, and Uttar Pradesh cluster together as the extreme high-risk group, with mean CMPR three times that of the low cluster. Cluster 2 — the low-CMPR group — is composed almost entirely of southern and small union territories, consistent with the LASSO finding that high female literacy in these states suppresses marriage rates. Cluster 3 forms a large heterogeneous middle group spanning central and northwestern states.

8.1.2. SC Female — 2011

By 2011, Cluster 1 shrank to only four states (Bihar, Jharkhand, Meghalaya, Rajasthan; mean CMPR = 29.3), while Himachal Pradesh, Mizoram, and Odisha migrated out to lower clusters:

Cluster	Mean CMPR	States
1	29.3	Bihar, Jharkhand, Meghalaya, Rajasthan
2	14.4	Dadra & Nagar Haveli, Daman & Diu, Goa, Jammu & Kashmir, Kerala, Maharashtra, Manipur, NCT of Delhi, Puducherry, Tamil Nadu, Tripura, Uttarakhand
3	18.5	Andhra Pradesh, Assam, Chandigarh, Chhattisgarh, Gujarat, Haryana, Himachal Pradesh, Karnataka, Madhya Pradesh, Mizoram, Odisha, Punjab, Sikkim, Uttar Pradesh, West Bengal

Table 12: SC Female Clusters, 2011 (mean CMPR at age_14_17)

Insight: Uttar Pradesh exited the highest-burden cluster between 2001 and 2011 — a notable improvement — moving to Cluster 3 (mean 18.5) alongside Odisha and Himachal Pradesh. Bihar, Jharkhand, and Rajasthan remained in Cluster 1 throughout, confirming their status as structurally resistant high-burden states for SC female child marriage. The mean CMPR of the high cluster fell from 36.9 to 29.3, reflecting overall progress but persistent concentration of burden.

8.1.3. SC Male — 2001 and 2011

Male CMPR levels are substantially lower. In 2001, Cluster 1 (mean 7.8) comprised the same north + east states; by 2011 the cluster structure reorganised with mean CMPRs of 4.6, 2.9, and 4.9, and a new cluster of just three states (Himachal Pradesh, Mizoram, Sikkim) at mean 4.9 — an unusual grouping that reflects these states' specific SC male marriage patterns not explained by the female literacy pathway.

8.2. ST Clustering

8.2.1. ST Female — 2001

Cluster	Mean CMPR	States
1	11.7	Arunachal Pradesh, Himachal Pradesh, Lakshadweep, Manipur, Mizoram, Nagaland, Sikkim, Uttaranchal
2	30.0	Bihar, Jammu & Kashmir, Jharkhand, Madhya Pradesh, Rajasthan, Orissa, Uttar Pradesh, West Bengal
3	21.2	Andaman & Nicobar, Andhra Pradesh, Assam, Chhattisgarh, Dadra & Nagar Haveli, Daman & Diu, Goa, Gujarat, Karnataka, Kerala, Maharashtra, Meghalaya, Tamil Nadu, Tripura

Table 13: ST Female Clusters, 2001 (mean CMPR at age_14_17)

Insight: The northeast states (Mizoram, Manipur, Nagaland, Arunachal Pradesh) and Lakshadweep cluster firmly in the low-CMPR group. The high-burden cluster is entirely composed of central and eastern mainland states — where the below-primary share (ST F) identified by LASSO is highest and school attendance was the operative protective mechanism in 2001.

8.2.2. ST Female — 2011

The cluster structure shifted substantially: Cluster 1 (mean 22.8) absorbed 17 states including previously-mid states like Tamil Nadu, Tripura, and Dadra & Nagar Haveli, indicating that the middle tier converged upward or failed to maintain previous gains. Cluster 2 (mean 10.7) shrank to only Himachal Pradesh, Nagaland, and Sikkim — a small, geographically dispersed low-burden outlier group.

Insight: The ST clustering between 2001 and 2011 shows less clear improvement than SC. The fact that more states joined the middle-to-high cluster by 2011 is consistent with the LASSO finding that the below-primary predictor barely attenuated ($+1.72 \rightarrow +1.66$): structural educational exclusion persisted, and dropout replaced attendance as the operative risk mechanism.

8.3. Hindu and Muslim Female Clustering

8.3.1. Hindu Female

In 2001, Cluster 1 (mean 27.3) was anchored by Bihar, Rajasthan, Madhya Pradesh, Jharkhand, Uttar Pradesh, and Chhattisgarh — the same high-literacy-deficit states

identified by LASSO. By 2011, Cluster 2 (mean 22.9, comprising only Arunachal Pradesh, Bihar, Rajasthan) separated as the extreme outlier; Bihar's extraordinarily high Hindu female CMPR drove it into its own small cluster. Cluster 1 (mean 14.4) expanded to 18 states in 2011, reflecting broad improvement across southern, western, and northeastern states.

8.3.2. Muslim Female

The 2001 Cluster 1 (mean 24.8) grouped 14 states including Assam, Meghalaya, Mizoram, Nagaland, and Jharkhand. By 2011, Cluster 1 shrank to 6 states (mean 26.0): Bihar, Haryana, Meghalaya, Mizoram, Nagaland, Rajasthan. This cluster is notable for its geographic diversity — Mizoram and Nagaland (northeast, Christian-majority) appearing alongside Haryana and Bihar in a high Muslim female CMPR group reflects the small Muslim population sizes in those states making CMPR volatile.

The persistence of Below primary (Muslim F) as the sole LASSO predictor in 2011 ($\beta = +2.71$) is reinforced by the clustering: the high-CMPR Muslim cluster is precisely the set of states where Muslim female educational attainment is truncated at below-primary level.

8.4. Christian Female Clustering

In 2001, Cluster 1 (mean 10.0) included most southern states, Mizoram, and UTs — the low-CMPR Christian heartland. Clusters 2 and 3 were close in mean (14.0 and 14.9), reflecting more heterogeneous patterns in northeast and eastern India.

By 2011, the structure shifted: Lakshadweep isolated into its own Cluster 3 with mean CMPR = 0.0 (effectively no Christian child marriage reported), while Cluster 2 (mean 8.3) absorbed most southern and northeastern states. Cluster 1 (mean 12.5) comprised 16 states including Jharkhand, Rajasthan, Bihar, and Nagaland.

The sign reversal of Below primary (Christian F) between 2001 (-4.39) and 2011 ($+1.41$) is reflected in the clustering: in 2001 the low-CMPR cluster (Cluster 1) corresponded to high-literacy southern states that also had relatively high below-primary shares among older generations, creating a spurious negative coefficient. By 2011, younger cohorts and geographic redistribution erased that pattern.

8.5. Total Female Clustering

Cluster	Mean CMPR	States (2001)
1	21.0	Chandigarh, Daman & Diu, Goa, Haryana, Himachal Pradesh, Maharashtra, Manipur, Mizoram, Punjab, Sikkim, Delhi, Uttaranchal, Tripura
2	29.3	Andhra Pradesh, Arunachal Pradesh, Assam, Chhattisgarh, Gujarat, Karnataka, Madhya Pradesh, Meghalaya, Nagaland, Orissa, West Bengal
3	35.3	Bihar, Dadra & Nagar Haveli, Jammu & Kashmir, Jharkhand, Rajasthan, Uttar Pradesh

Table 14: Total Female Clusters, 2001 (mean CMPR at age_14_17)

By 2011 the high-burden cluster (mean 23.7) comprises Bihar, Jharkhand, Rajasthan, Uttar Pradesh, Jammu & Kashmir, and Arunachal Pradesh — exactly the states that the LASSO total female model in 2001 best explained via literacy deficits ($\beta = -4.90$). The mean CMPR of the highest cluster fell from 35.3 to 23.7, a meaningful absolute decline, while the middle cluster (mean 21.1) includes Haryana, Gujarat, Manipur, and Punjab — states where economic development has not yet translated fully into reduced child marriage.

8.6. Cross-Group Clustering (2011)

The cross-group clustering using all six female CMPR dimensions simultaneously produced three clusters:

Cluster	States
1	Andhra Pradesh, Bihar, Jharkhand, Madhya Pradesh, Rajasthan, Tripura, West Bengal
2	Daman & Diu, Goa, Kerala, Manipur
3	Assam, Chhattisgarh, Dadra & Nagar Haveli, Gujarat, Himachal Pradesh, Jammu & Kashmir, Karnataka, Maharashtra, Meghalaya, Mizoram, Odisha, Sikkim, Tamil Nadu, Uttar Pradesh, Uttarakhand

Table 15: Cross-Group Female CMPR Clusters, 2011

Insight: The cross-group heatmap (page 1 of `all_clustering_graphs.pdf`) is the most

revealing visualisation in the clustering output. Cluster 1 states are uniformly high-CMPR across *all* religious and caste groups simultaneously: Bihar shows +2.4 SD for SC female, +2.3 SD for Muslim female, and +1.4 SD for Hindu female in the same row. This means these states are not high-burden in just one community — they are structurally high-burden across the board, indicating a state-level environment (governance, enforcement, poverty, educational infrastructure) rather than community-specific practices as the dominant driver. Cluster 2 — just four states — is the low-CMPR outlier group across all dimensions, dominated by southern and small UTs. Cluster 3 is the large heterogeneous middle.

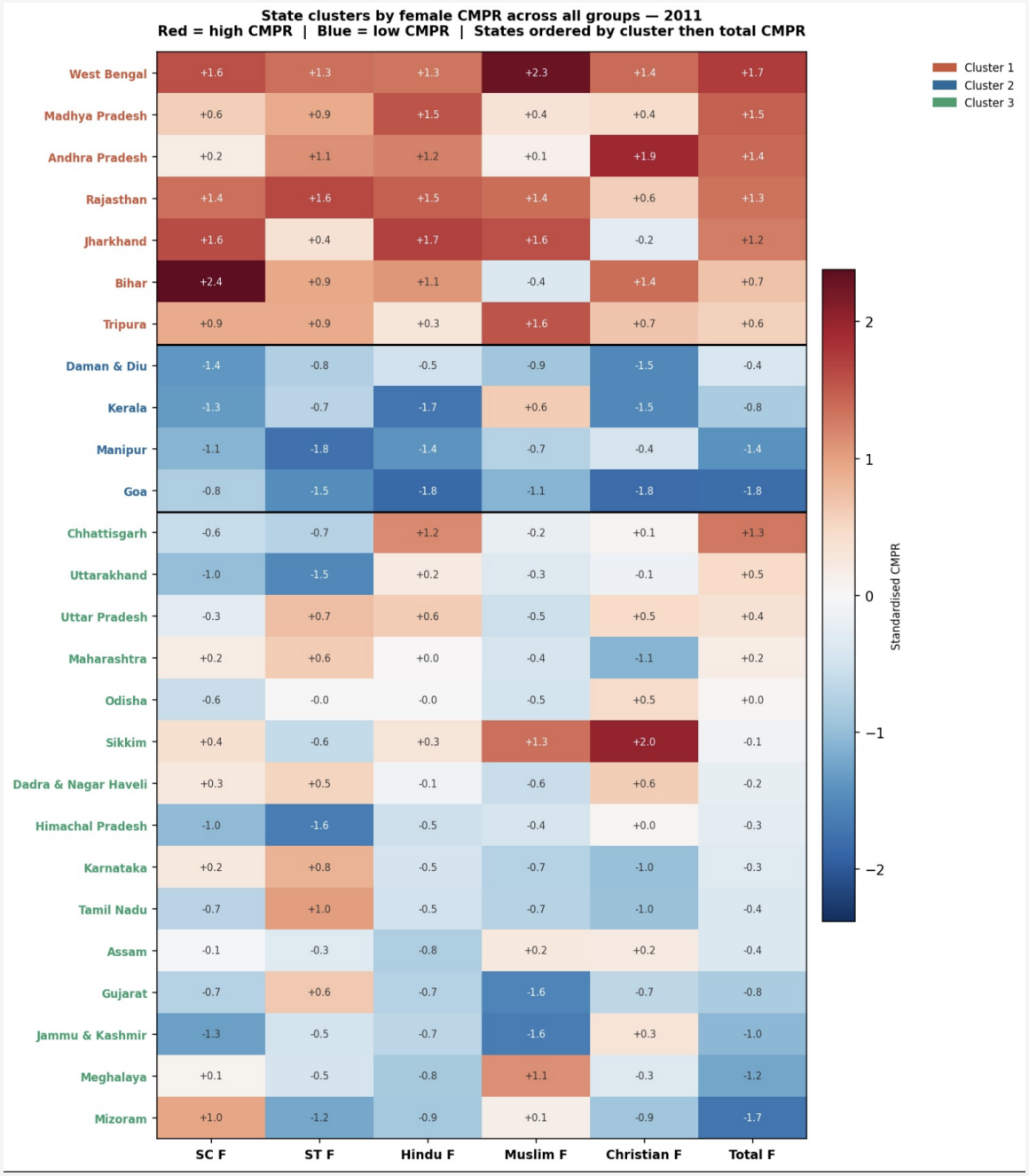

Figure 8: Cross-Group Female CMPR Heatmap, 2011

Explanation: It shows standardised CMPR across six group-gender columns (SC F, ST F, Hindu F, Muslim F, Christian F, Total F) for all 35 states, ordered by cluster. The deep red of West Bengal (+2.3 Muslim F, +1.3 ST F, +1.7 Total F) and Bihar (+2.4 SC F) versus the solid blue of Goa and Kerala across every column makes the "universally high-burden vs. universally low-burden" distinction viscerally clear.

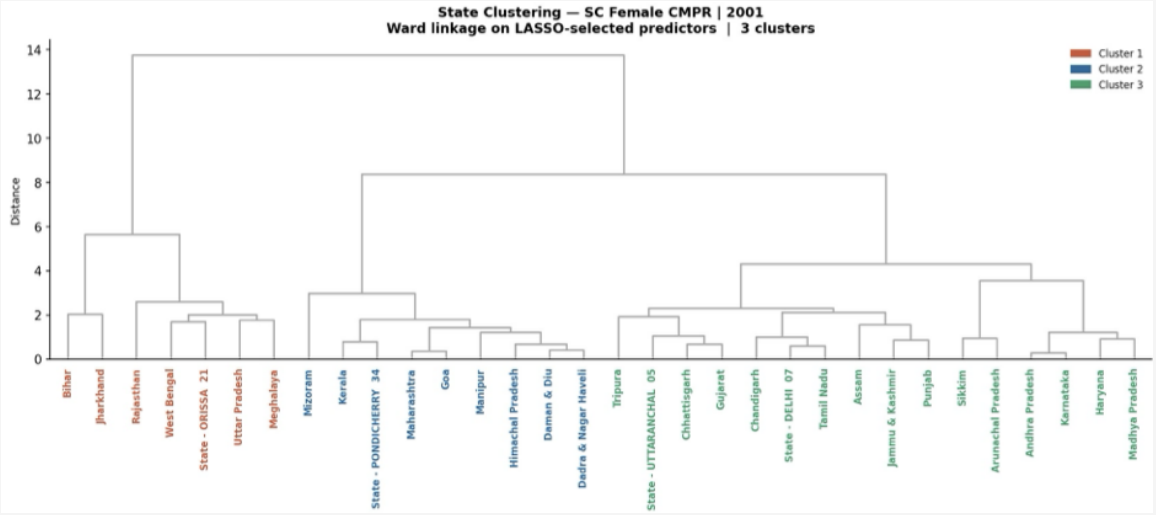

Figure 9: State Clustering Dendrogram: SC Female, 2001

Explanation: The dendrogram shows the Ward-linkage merge distances, making explicit how far apart the three clusters are. The large gap between the branch linking Bihar–Jharkhand and the rest of the states ($\approx$ distance 14) versus the tight sub-clusters within each group indicates that the high-burden states form a genuinely discrete group, not just the top of a smooth distribution.

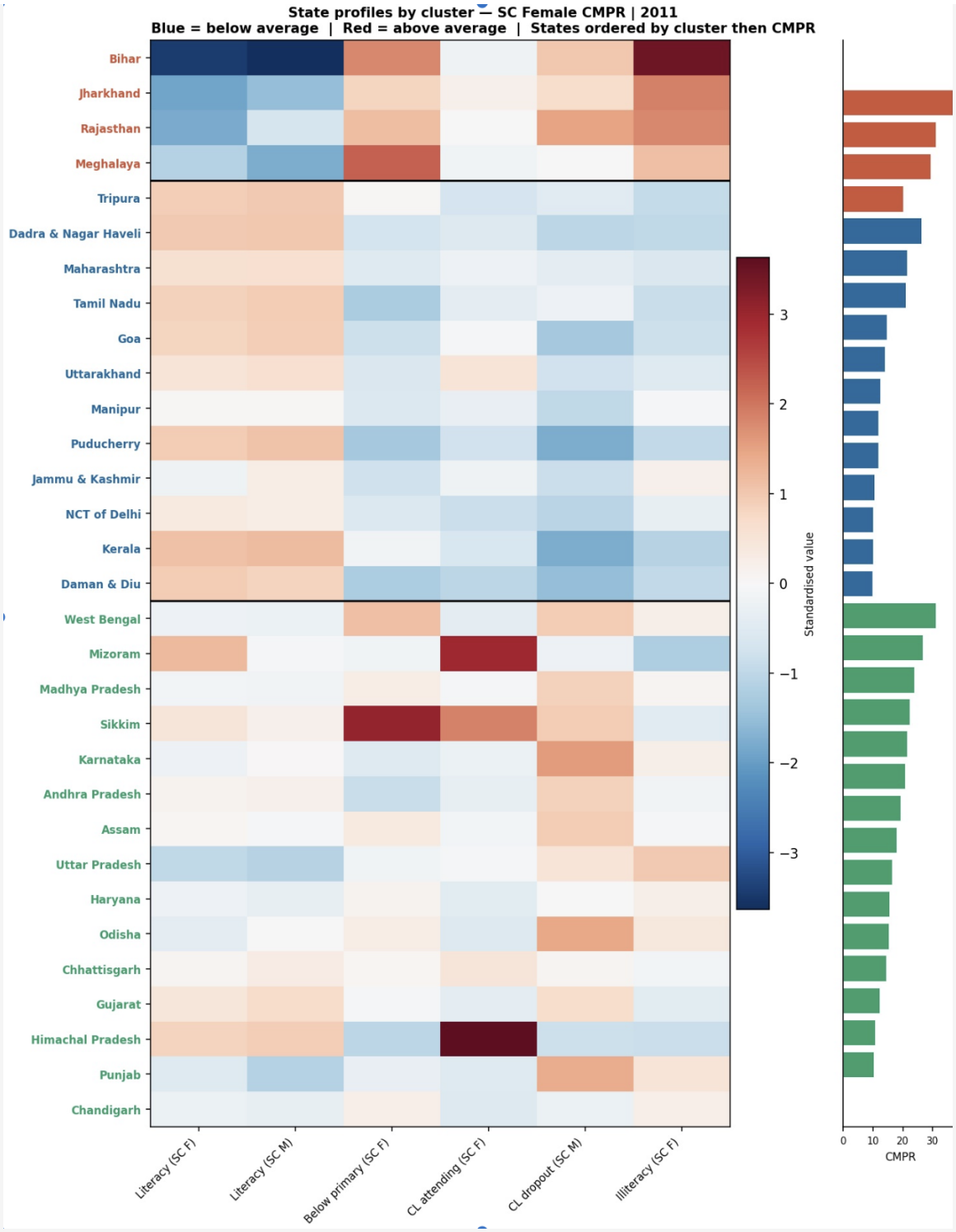

Figure 10: State Profiles Heatmap: SC Female 2011

Explanation: This heatmap shows the six LASSO-selected predictor values alongside the CMPR bar chart for each state, colour-coded by cluster. It directly connects the LASSO findings (Section 7) with the clustering (Section 8): Cluster 1 states (Bihar, Jharkhand, Rajasthan, Meghalaya) are visibly high on Below primary (SC F) and

low on Literacy (SC F), while Cluster 2 shows the opposite pattern. The CL dropout (SC M) column, new in 2011, is visibly elevated in the high-CMPR cluster, providing visual confirmation of Finding 2.

9. Project Dashboard

Link to Dashboard (and blog): https://project-bachpan.vercel.app

9.1. Overview and Purpose

An interactive web dashboard has been developed to allow non-technical users — including NGO staff and policy researchers — to explore the CMPR indexes across states, age brackets, and sub-groups without requiring access to raw data or code.

9.2. Dashboard Snippet

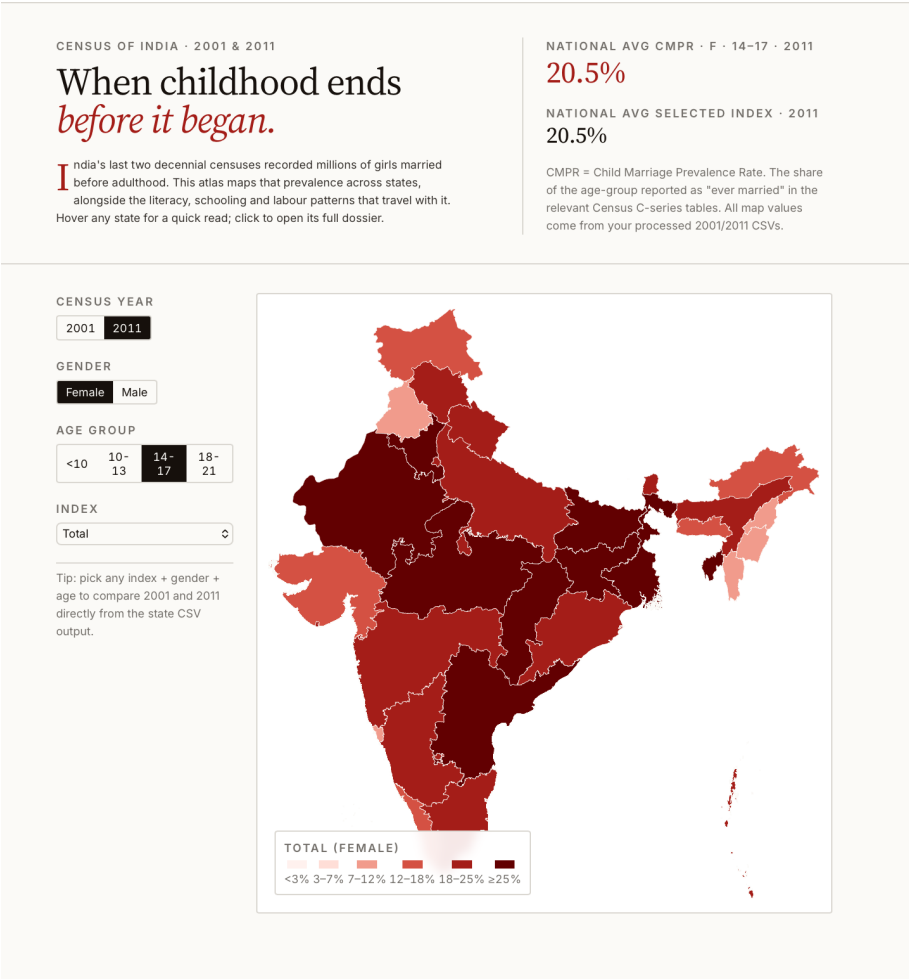

Figure 11: Interactive Dashboard

9.3. Dashboard Features

The dashboard provides the following views and controls:

- **State-level choropleth map**: India map coloured by CMPR at a user-selected age bracket, with controls to switch between total, SC, ST, and religion sub-groups.
- **2001 vs. 2011 comparison**: Side-by-side state rankings or bar charts showing CMPR in both census years for the selected group and bracket.
- **Sub-group toggle**: Switch the active dataset between general population, SC, ST, Hindu, Muslim, Christian, Sikh, Buddhist, and Jain views.
- **Regression results explorer**: Visualisation of LASSO coefficients for each group and year, showing which predictors were selected and their relative magnitudes.
- **Feature correlation explorer**: Scatter plots of any two indexes across all 35 states, with the active bracket selectable.
- **Downloadable filtered CSV**: Export the currently displayed state-bracket-group slice for external analysis.

10. Known Limitations and Caveats

1. **Age band mismatch.** The four canonical age brackets do not align perfectly with every census table's reporting bands. C-02's 5-year bands introduce approximation errors, most notably that `age_10_13` includes age 14 and `age_14_17` excludes it.
2. **Partial `age_18_21` coverage.** In C-08, C-09, and C-12, ages 20–21 are embedded in the "20–24" band and cannot be extracted. The `age_18_21` bracket for these sources therefore captures only ages 18 and 19.
3. **C-12 age floor.** The school attendance dataset begins at age 5; the `age_below10` bracket covers only ages 5–9, not 0–4.
4. **Small sample size.** With 35 state/UT observations per bracket, all models operate in a small-n regime. LOO-CV mitigates overfitting, but the CV R^2 values — especially for religion models — should be interpreted cautiously. The SC male 2011 model (11 predictors, CV $R^2 = 0.166$) in particular shows signs of overfitting; its predictor list should not be taken as a reliable set of causal drivers.
5. **State-level aggregation.** All indexes are computed at the state level. District-level variation, which may be substantial, is not captured.
6. **Cross-sectional data within each year.** The 2001 and 2011 datasets are each cross-sectional snapshots. The year-on-year comparison of coefficients approximates a longitudinal design but does not constitute panel estimation in the strict sense; unobserved time-invariant state characteristics are not controlled for.

7. **Underreporting of marriage.** Child marriage is illegal; some incidence is likely underreported in official census responses, meaning our CMPR indexes provide lower bounds.

8. **CMPR as a proxy.** "Currently married" is a proxy for child marriage, not a direct measure. It captures individuals married at any age who are currently in that marital state, not necessarily those who married as children.

9. **Religion predictors limited to education.** The `df_religion` predictor set is restricted to literacy and school-level education variables from C-09. Occupation, income, and urbanisation — which likely differ substantially across religious communities — are not available at the religion × state level in any census sub-table used here. This is the most likely explanation for the sharp CV R^2 declines in religion models between 2001 and 2011.

10. **Hindu male 2001 model.** The LASSO model for `CMPR_hindu_male` in 2001 selected zero predictors, with α at its upper bound of 100 and $R^2_{\text{cv}} = -0.06$. This reflects near-zero explainable variance in Hindu male CMPR in 2001 using the available predictors, not a model error. Results for this group-year should not be interpreted.

11. **Christian Below primary sign reversal.** The sign reversal of Below primary (Christian F) from −4.39 (2001) to +1.41 (2011) is interpreted as a composition effect but cannot be confirmed without district-level or longitudinal data. The 2001 coefficient may partly reflect spatial confounding between Christian community location (southern, low-CMPR states) and educational attainment.

11. Conclusion

This project constructs a comprehensive, multi-dimensional dataset of child marriage rates and their correlates across all Indian states, drawing on 15+ census sub-tables from the 2001 and 2011 waves. By computing CMPR and associated structural indexes stratified by sex, scheduled status, and religious community, the project produces a state-level analytical resource enabling both descriptive comparison and formal regression modelling.

The LASSO analysis — verified against `lasso_log_20260426_181518.txt` — and the hierarchical clustering — verified against `clustering_log_20260426_182511.txt` — together yield six substantive findings:

1. **Female literacy is the universal protective driver**: it appears in every model across all groups and both genders, with the largest coefficient magnitudes (−4.90 for Total female in 2001).

2. **SC and ST models show a parallel attendance-to-dropout shift**: school attendance (even under child labour) was protective in 2001; by 2011 male child-labour dropout replaced it as the dominant risk signal, pointing to deepened household-level economic distress.

3. **ST models are the best-fitting models in the study** (CV $R^2 = 0.711$ for ST female 2001), and below-primary share of ST females is the most temporally stable predictor ($+1.72 \rightarrow +1.66$), confirming structurally entrenched educational exclusion.

4. **Religion models show declining explanatory power in 2011**, converging to a single predictor each: female literacy for Hindus, below-primary share for Muslims, and an anomalous sign-reversing below-primary term for Christians.

5. **Universal coefficient attenuation** from 2001 to 2011 (e.g. Literacy SC F: $-4.31 \rightarrow -0.72$) reflects both genuine progress and structural changes that have made single-factor explanations less adequate.

6. **Cross-group clustering confirms that the highest-burden states — Bihar, Jharkhand, Rajasthan — are universally high across all religious and caste groups**, ruling out community-specific practices as the primary explanation and implicating state-level structural factors.

The analysis is complemented by an interactive dashboard (`http://10.1.33.9:8081`) that visualises CMPR patterns across states, brackets, and sub-groups for non-technical audiences, and exposes the LASSO coefficient comparisons and clustering results as an explorable interface.

A. Appendix: Complete Index Reference

A.1. df_total Indexes (23 metrics × 2 sexes where applicable)

Table 16: Complete `df_total` index list

Column Name	Source Table
`CMPR_total_male`	C-02
`CMPR_total_female`	C-02
`CMPR_illiterate_male`	C-06
`CMPR_illiterate_female`	C-06
`CMPR_below_primary_male`	C-06
`CMPR_below_primary_female`	C-06
`CMPR_primary_male`	C-06

Column Name	Source Table
CMPR_primary_female	C-06
CMPR_middle_male	C-06
CMPR_middle_female	C-06
CMPR_matric_male	C-06
CMPR_matric_female	C-06
CMPR_main_workers_male	C-07
CMPR_main_workers_female	C-07
CMPR_cultivators_male	C-07
CMPR_cultivators_female	C-07
CMPR_agri_labourers_male	C-07
CMPR_agri_labourers_female	C-07
CMPR_household_industry_male	C-07
CMPR_other_workers_male	C-07
CMPR_other_workers_female	C-07
CMPR_non_workers_male	C-07
CMPR_non_workers_female	C-07

A.2. Bracket Map Reference

Bracket	C-02	C-08/09/12	C-06/07 (age at marriage)
age_below10	0–9	0–6, 7, 8, 9	Less than 10
age_10_13	10–14†	10, 11, 12, 13	10–11, 12–13
age_14_17	15–19†	14, 15, 16, 17	14–15, 16–17
age_18_21	20–24†	18, 19‡	18–19, 20–21

† 5-year band approximation. ‡ Ages 20–21 inseparable from 20–24 band.

Table 17: Age label sets by bracket and source table

A.3. LASSO Script: Key Implementation Details

The `cmpr_lasso_analysis.py` script implements the full modelling pipeline. Key implementation decisions:

- **Alpha grid**: 100 values log-spaced between 10^{-3} and 10^{2}, searched via `sklearn.linear_model.Lasso`
- **Convergence**: `max_iter=20000` to ensure convergence on the standardised feature space.

- **CV**: Leave-One-Out via `sklearn.model_selection.LeaveOneOut` for all groups with $n < 40$.
- **CV R^2 averaging**: `np.nanmean` used over LOO fold scores to handle rare negative folds from unstable held-out predictions without distorting the mean.
- **Alpha ceiling warning**: If the selected $\alpha = 100$ (upper grid bound), a warning is printed and the model is flagged as uninformative for that group-year. This occurred for Hindu male 2001 only.
- **Religion 2001 CMPR**: Derived via `compute_religion_cmpr_2001()` from the raw C-03 Appendix CSV, mapping "Less than 18" to `age_14_17`.
- **Outputs**: Per-group coefficient comparison charts, change-summary charts, regularisation path plots, a cross-dataset coefficient heatmap, and a summary CSV are saved to `regression_outputs/lasso/`.

A.4. Clustering Script: Key Implementation Details

The `cmpr_clustering.py` script implements Ward-linkage hierarchical clustering on LASSO-selected predictor features for each group and census year. Key details:

- **Linkage**: Ward's minimum variance criterion, which tends to produce compact, evenly-sized clusters appropriate for state-level geographic data.
- **Number of clusters**: $k = 3$ for all groups, confirmed by elbow curves (within-cluster sum of squares plotted for $k = 2$ to $k = 6$). All elbow curves show a clear bend at $k = 3$.
- **Features**: LASSO-selected predictors for each group (e.g. SC female clustering uses Literacy SC F, CL attending SC F, Below primary SC F, Literacy SC M for 2001).
- **Outputs**: Elbow curves, dendrograms, state profile heatmaps, and a cross-group heatmap saved to `regression_outputs/clustering/`. A cluster profile CSV is also generated.
- **Cross-group clustering**: A single clustering using all six group-gender CMPR columns simultaneously is run for 2011 to identify states that are uniformly high or low across all communities.

References

[1] Alice Douglass et al. Exploring the socioeconomic factors associated with girl child marriage in india. *International Journal of Environmental Research and Public Health*, 19:4567, 2022.

[2] Chang-Ting Hsieh et al. India's growth in the 2000s: Four facts. *CEPR VoxEU*, 2011.

[3] Registrar General & Census Commissioner, India. Census of india 2011: Literacy and education, 2011. Provisional data on literacy rates and gender gaps; available via Census of India publications.

[4] UNFPA India. Child marriage in india: Key insights from the nfhs-5, 2022.

[5] World Bank. Gdp growth (annual %) - india, 2026. World Development Indicators, Open Data.